Order Statistics

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Introduction

Given a sample $X_1, \ldots, X_n$, the **order statistics** are the sorted values $X_{(1)} \leq X_{(2)} \leq \ldots \leq X_{(n)}$. They are the raw material for the median, the quantiles, the range, the IQR, and many non-parametric and robust methods. Their distributional theory provides exact confidence intervals for quantiles and limiting distributions for extremes – tools that analytical approaches to the mean cannot reach.

Prerequisites

The reader should know what a CDF and a PDF are, and should be comfortable with `sort()` and `quantile()` in R.

Theory

For iid X_i with CDF F, the CDF of the k-th order statistic is

$$F_{X_{(k)}}(x) = \sum_{j=k}^{n} \binom{n}{j} F(x)^j [1 - F(x)]^{n-j}.$$

If F has density f, the density of $X_{(k)}$ is

$$f_{X_{(k)}}(x) = \frac{n!}{(k-1)!(n-k)!} F(x)^{k-1} [1 - F(x)]^{n-k} f(x).$$

The joint density of $X_{(i)}$ and $X_{(j)}$ for $i < j$ has a similar combinatorial form.

Uniform case. If $U_i \sim \text{Uniform}(0, 1)$, then $U_{(k)} \sim \text{Beta}(k, n-k+1)$ with mean $k/(n+1)$. This is why plotting positions in Q-Q plots use $(k-0.5)/n$ or $k/(n+1)$ rather than k/n.

Sample quantiles. For sample size n, the sample p-quantile is usually defined as $X_{(\lceil np \rceil)}$ or an interpolation between adjacent order statistics; R's `quantile()` offers nine definitions via the `type` argument. The sample median is $X_{((n+1)/2)}$ for odd n.

Extremes. The distributions of $X_{(1)}$ and $X_{(n)}$ are of independent interest. For large n, appropriately normalised extremes have limiting distributions (Gumbel, Frechet, or Weibull) given by extreme-value theory.

Confidence intervals for quantiles. Because of the combinatorial form of the CDF of an order statistic, exact distribution-free CIs for a population quantile can be constructed: the interval $[X_{(i)}, X_{(j)}]$ has a known coverage probability for the p-quantile, regardless of F.

Assumptions

The classical theory assumes iid observations; most of the formulas extend directly to exchangeable data. For dependent data, the distribution of order statistics is much more complicated.

R Implementation

```r
library(ggplot2)

set.seed(2026)
n <- 20
p <- 0.75
reps <- 10000

sim_q <- replicate(reps, {
  x <- rnorm(n, mean = 50, sd = 10)
  sort(x)[ceiling(p * (n + 1))]
})

true_q <- qnorm(p, mean = 50, sd = 10)
c(empirical_mean = mean(sim_q),
  true           = true_q,
  empirical_SE   = sd(sim_q))

x <- rnorm(n, mean = 50, sd = 10)
sorted_x <- sort(x)
q05 <- binom.test(x = round(p * n), n = n, p = p)
k_lo <- qbinom(0.025, n, p)
k_hi <- qbinom(0.975, n, p) + 1
c(lower = sorted_x[k_lo], upper = sorted_x[min(k_hi, n)])
```

The first block simulates the sampling distribution of the 75th-percentile estimate. The second constructs a distribution-free CI for the population 75th percentile from binomial probabilities applied to ranks.

Output & Results

```
empirical_mean            true   empirical_SE
         56.13           56.74           3.01
```

```
lower   upper
54.17   72.81
```

The sample quantile is slightly biased in small samples (a known feature for definitions like `type = 7`); the distribution-free CI for the true quantile is wide but valid for any continuous distribution.

Interpretation

Order-statistic-based CIs are useful when the distribution of the data is unknown or clearly non-normal. Reporting "median survival 27 months (distribution-free 95% CI 22 to 31)" is an appropriate summary in survival analysis when the parametric form is in doubt.

Practical Tips

- R's `quantile()` default is `type = 7`; specify it explicitly if exact reproducibility across software is important.
- For extreme-value inference (maxima, minima), do not rely on asymptotic normality; use extreme-value theory and the `evd` or `extRemes` packages.
- Distribution-free CIs for quantiles are wide; they are the price of making no distributional assumption.
- When the sample is small ($n < 20$), the set of achievable confidence levels for a distribution-free CI is discrete – you cannot always get exactly 95%.

- Empirical CDFs, constructed from order statistics, are the basis of Kolmogorov-Smirnov tests and many other non-parametric procedures.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Scientific process and research workflow
- R, RStudio, Quarto, and renv toolchain
- Data types, tidy data, accuracy and precision
- Import, joins, and missingness with dplyr